

vSphere Export Guide

This guide covers hyper2kvm's direct export capabilities for VMware vSphere environments.

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Overview

hyper2kvm provides multiple methods to export VMs from vSphere environments:

1. **Direct Export** - Convert VMware disks directly to QCOW2/RAW using internal converters
2. **HTTP Download** - Download VM folders via vSphere's HTTP API
3. **VDDK Download** - Pull individual disks using VMware's VDDK library
4. **OVF/OVA Export** - Export as standard OVF or OVA packages

All export modes support: - Snapshot management (create, use existing, no snapshot) - Credential resolution (environment, config file, interactive) - Progress tracking and logging - Error recovery and retry logic

Export Modes

1. Direct Export (`export_mode="export"`)

Converts VMware disks directly to KVM-compatible formats using hyper2kvm's internal converters.

Features: - Guest OS detection and modification - Filesystem fixes (fstab, network config)
- Initramfs regeneration - Driver injection

Use Case: Full V2V migration with guest modifications

Example:

```
from hyper2kvm.vmware.clients.client import VMwareClient, ExportOptions

client = VMwareClient(
    vcenter_host='vcenter.example.com',
    username='administrator@vsphere.local',
    password='password'
)

await client.async_export_vm(
    vm_name='production-web',
    output_dir='/data/exports',
    export_mode='export' # Direct export with conversion
)
```

2. HTTP Download (`export_mode="download_only"`)

Downloads exact VM folder contents from the datastore using vSphere's `/folder` HTTP API.

Features: - Preserves original VMware formats - Downloads all VM files (VMDK, VMX, etc.) - Fast transfer via HTTPS - No conversion overhead

Use Case: Backup, archival, or manual conversion later

Example:

```
options = ExportOptions(
    vm_name='backup-vm',
    output_dir='/backups',
    export_mode='download_only' # HTTP download, no conversion
)
```

```
)  
  
await client.export_vm(options)
```

3. VDDK Single-Disk (`export_mode="vddk_download"`)

Pulls a single disk as raw bytes using VMware VDDK.

Features: - Fast binary transfer - Block-level access - Efficient for single disks - Requires VDDK library

Use Case: Extract specific disks without full VM download

Example:

```
options = ExportOptions(  
    vm_name='database-vm',  
    output_dir='/data/disks',  
    export_mode='vddk_download',  
    disk_index=0 # First disk only  
)  
  
await client.export_vm(options)
```

4. OVF Export (`export_mode="ovf_export"`)

Exports VM as standard OVF format using VMware APIs.

Use Case: Portable VM format for import to other platforms

5. OVA Export (`export_mode="ova_export"`)

Exports VM as single OVA archive file.

Use Case: Simplified distribution and portability

ExportOptions API

The `ExportOptions` class configures export behavior:

```
from hyper2kvm.vmware.clients.client import ExportOptions  
  
options = ExportOptions(  
    # ...  
)
```

```

# Required
vm_name='target-vm',
output_dir='/exports',

# Export mode
export_mode='export', # 'export', 'download_only', 'vddk_download',
                        'ovf_export', 'ova_export'

# Snapshot handling
snapshot_mode='create', # 'create', 'use_existing', 'none'
snapshot_name='migration-snapshot',

# Conversion options (for export mode)
output_format='qcow2', # 'qcow2', 'raw'
compress=True,

# Authentication
vcenter_host='vcenter.example.com',
username='admin@vsphere.local',
password_file='/secure/vcpass.txt',

# Performance
parallel_downloads=True,
max_workers=4,

# Advanced
extra_args=['--verbose'],
timeout=3600,
)

```

Key Parameters

Export Mode

- `export` - Full conversion with guest modifications
- `download_only` - Raw file download via HTTP
- `vddk_download` - Single disk via VDDK
- `ovf_export` - Standard OVF format
- `ova_export` - Single OVA archive

Snapshot Mode

- `create` - Create new snapshot for export
- `use_existing` - Use existing snapshot by name

- `none` - Export from current state (downtime required)

Output Format

- `qcow2` - QEMU Copy-On-Write (recommended)
- `raw` - Raw disk image
- `vdi` - VirtualBox format

Export Workflows

Basic Export Workflow

```
import asyncio
from hyper2kvm.vmware.clients.client import VMwareClient

async def export_vm():
    client = VMwareClient(
        vcenter_host='vcenter.example.com',
        username='admin@vsphere.local',
        password='password'
    )

    await client.async_export_vm(
        vm_name='web-server-01',
        output_dir='/data/migrations',
        export_mode='export'
    )

asyncio.run(export_vm())
```

Batch Export with Progress

```
from hyper2kvm.manifest.batch_progress import ProgressTracker

async def batch_export(vm_list):
    client = VMwareClient(...)
    tracker = ProgressTracker('/tmp/progress.json', 'batch-export',
                              len(vm_list))

    for vm_name in vm_list:
        tracker.start_vm(vm_name)
        tracker.update_vm_stage(vm_name, 'export')
```

```

    try:
        await client.async_export_vm(
            vm_name=vm_name,
            output_dir=f'/exports/{vm_name}',
            export_mode='export'
        )
        tracker.complete_vm(vm_name, success=True)
    except Exception as e:
        tracker.complete_vm(vm_name, success=False, error=str(e))

tracker.cleanup()

```

Export with Snapshot Management

```

# Create snapshot for export
options = ExportOptions(
    vm_name='production-db',
    output_dir='/backups',
    export_mode='download_only',
    snapshot_mode='create',
    snapshot_name='pre-migration-backup',
    snapshot_description='Snapshot for migration testing',
    snapshot_memory=False # Faster, no RAM state
)

await client.export_vm(options)

```

VDDK vs HTTP Download

Use VDDK When:

- Exporting single disks
- Need block-level access
- Want fastest transfer speeds
- Have VDDK library installed

Use HTTP When:

- Need complete VM folder
- Want all VM metadata
- No VDDK library available

- Firewall restricts VDDK ports

Comparison

Feature	VDDK	HTTP Download
Speed	Fastest	Fast
Files	Single disk	All VM files
Requirements	VDDK library	HTTPS only
Ports	902 (NBD)	443 (HTTPS)
Metadata	No	Yes (VMX, NVRAM)

Authentication

Environment Variables

```
export VSPHERE_HOST='vcenter.example.com'
export VSPHERE_USER='admin@vsphere.local'
export VSPHERE_PASSWORD='password'

python -m hyper2kvm ...
```

Password File

```
options = ExportOptions(
    vcenter_host='vcenter.example.com',
    username='admin@vsphere.local',
    password_file='/secure/.vcpass', # One line: password
    ...
)
```

Config File

```
# config.yaml
cmd: vsphere
vs_action: export
vcenter_host: vcenter.example.com
vcenter_user: admin@vsphere.local
```

```
vc_password_env: VC_PASSWORD
```

```
export_mode: export
```

```
output_dir: /data/exports
```

```
hyper2kvm --config config.yaml --vm-name web-server-01
```

Best Practices

1. Use Snapshots for Live VMs

Always create snapshots when exporting running VMs:

```
export_mode='export',  
snapshot_mode='create',  
snapshot_name=f'migration-{timestamp}',  
snapshot_memory=False # Faster
```

2. Plan Bandwidth Usage

For large VMs, schedule exports during off-peak hours:

```
# Limit parallel downloads  
max_workers=2 # Instead of default 4  
  
# Use compression  
compress=True
```

3. Validate Exports

Always validate exported disks:

```
from hyper2kvm.validation import DiskValidator  
  
validator = DiskValidator()  
report = validator.validate({  
    'output_path': '/exports/vm-disk.qcow2',  
    'format': 'qcow2',  
    'minimum_size': 10 * 1024 * 1024 * 1024 # 10 GB  
})
```

```
if report.has_errors():  
    print("Validation failed!")
```

4. Clean Up Snapshots

Remove temporary snapshots after export:

```
await client.async_delete_snapshot(  
    vm_name='production-db',  
    snapshot_name='migration-snapshot'  
)
```

5. Monitor Progress

Use progress tracking for visibility:

```
from hyper2kvm.manifest.batch_progress import ProgressTracker  
  
tracker = ProgressTracker(  
    progress_file='/tmp/export-progress.json',  
    batch_id='nightly-backup',  
    total_vms=50  
)
```

Troubleshooting

Connection Issues

Error: Connection refused or TLS handshake failed

Solution:

```
# Verify connectivity  
curl -k https://vcenter.example.com/  
  
# Check certificate  
export VSPHERE_VERIFY_SSL=false # For testing only  
  
# Check credentials  
export VSPHERE_DEBUG=1
```

Slow Export Performance

Issue: Export taking too long

Solutions: 1. Use `export_mode='download_only'` for raw files 2. Increase `max_workers` for parallel downloads 3. Use VDDK instead of HTTP for single disks 4. Check network bandwidth between hyper2kvm host and vSphere

Snapshot Failures

Error: Snapshot creation failed

Solutions: 1. Verify VM has no existing snapshot locks 2. Check datastore has sufficient space 3. Use `snapshot_memory=False` to avoid RAM dump 4. Try `snapshot_mode='none'` with VM powered off

VDDK Not Found

Error: VDDK library not found

Solution:

```
# Install VDDK
wget https://developer.vmware.com/downloads/.../VMware-vix-disklib-*.tar.gz
tar xzf VMware-vix-disklib-*.tar.gz
export LD_LIBRARY_PATH=/path/to/vmware-vix-disklib/lib64:$LD_LIBRARY_PATH
```

Permission Denied

Error: Permission denied during export

Solution: 1. Verify user has VM export permissions in vSphere 2. Check datastore accessibility 3. Ensure snapshot creation rights 4. Review vCenter role assignments

See Also

- [vSphere Design](#) - Architecture details
- [Batch Migration Guide](#) - Batch processing
- [Library API](#) - Programmatic usage
- [Troubleshooting](#) - Common issues

Examples

Complete examples are available in: - `examples/yaml/` - YAML configuration samples - `examples/json/` - JSON manifest examples - `test-confs/` - Test configurations

For more information, see the main [documentation index](#).